INCH-POUND
MIL-C-17361/1A(SH)
13 March 1995
SUPERSEDING
MIL-C-17361/1(SH)
10 March 1989

MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB - A50, AQB - A50D, AQB - A51, AND NQB - A50

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-17361.

AMSC N/A
FSC 5925
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.

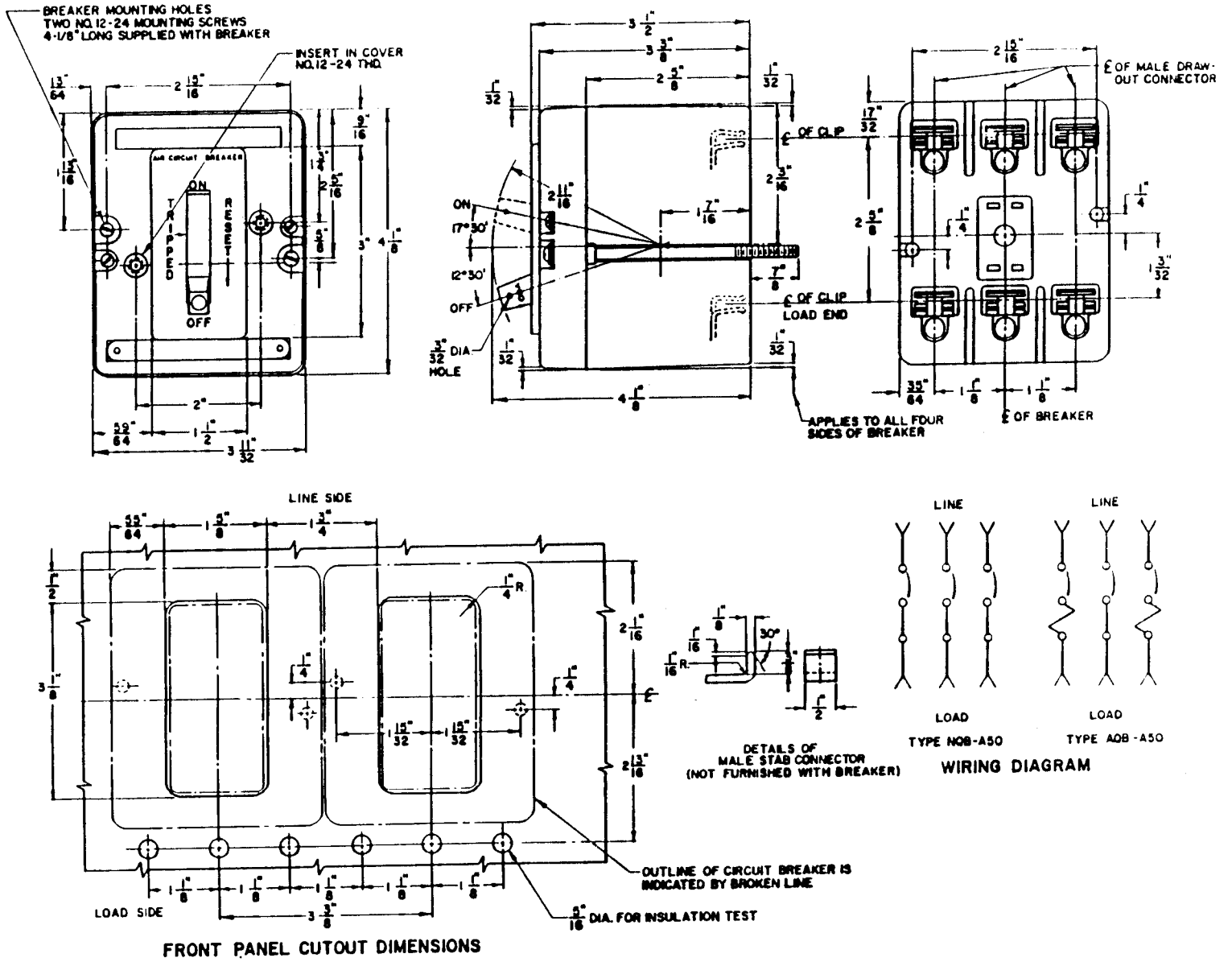


FIGURE 1. Circuit breaker type AQB-A50, AQB-A50D, AQB-A51, and NQB-A50 outline and mounting dimensions

Source: https://assist.dla.mil -- Downloaded: 202610-18T10:00Z
Check the source to verify that this is the current version before use.

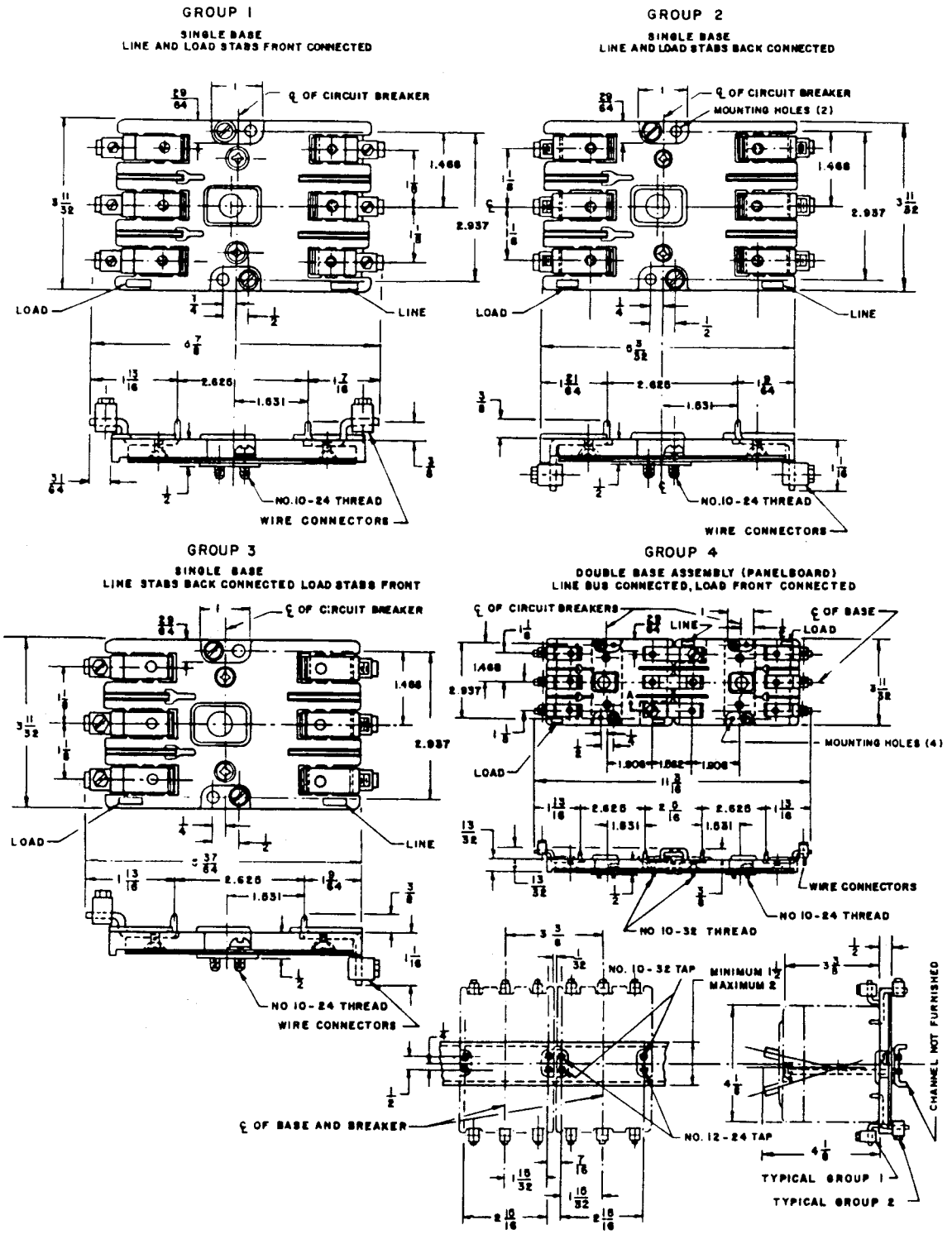


FIGURE 2. Mounting bases for circuit breaker type AQB-A50, AQB-A50D, AQB-A51, and NQB-A50 outline and mounting dimensions

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.

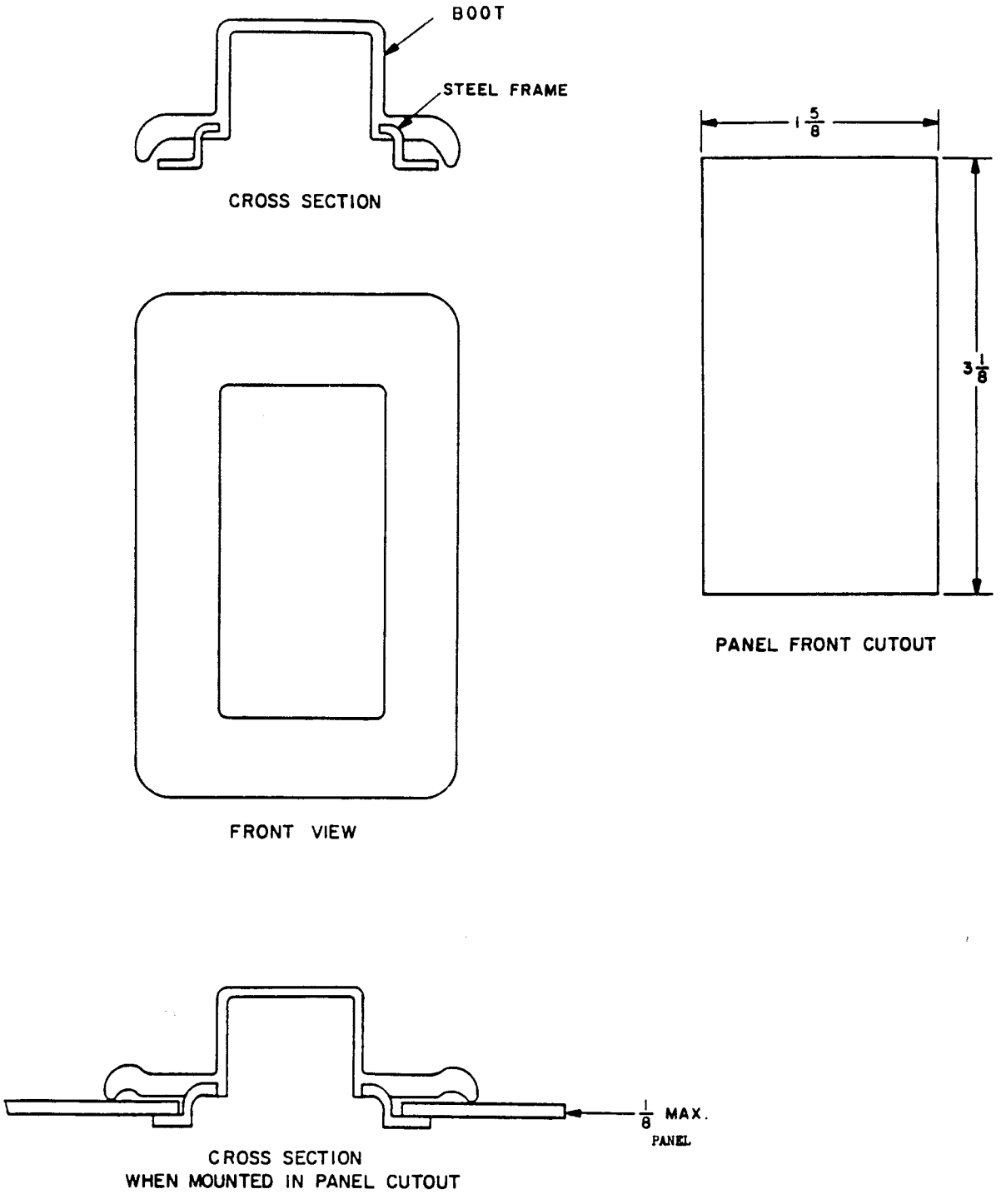


FIGURE 3. Circuit breaker handle boot for AQB-A50, AQB-A50D, AQB-A51, and NQB-A50

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions: See figures 1 and 2.
Handle boot: See figure 3 and table I.
Maximum rated continuous current: 50 amperes.

		Applicability		
		A50	A50D	A51
Voltage and frequency:	500 Vac, 60 Hz	X		X
	500 Vac, 400 Hz	X		
	250 Vdc	X	X	
Interrupting rating:	12,000 amps dc		X	
	5,000 amps 60 Hz	X		X
	2,500 amps, 400 Hz	X		
	2,500 amps dc	X		

Peak let-through current: Not applicable.
Trip unit: Noninterchangeable.
Trip unit rating (amperes): See table II.
LTD calibration: See table III.
Instantaneous calibration: See table II and IIa.
Current limiting fuse: Not applicable.
Type of connections: Back (removable mtg.).
Electric operator: Not available.
Undervoltage trip or shunt trip: Neither available.
Auxiliary switch: Not available.
Handle lock: Group 1. (Figure 1 of MIL-C-17361)
Additional identification: Front cover - "ON-OFF".
Frame size - ampere rating.
Additional mounting requirements: Each circuit breaker shall have two threaded metal inserts molded in the cover for insertion of the supporting screws. Stud terminals or lug terminals connected to the circuit breaker shall not turn to reduce the electrical clearances. Lug terminals shall be supplied with mounting bases.
Shock test: Mounting shall be in a vertical position in accordance with fixture 6E of MIL-S-901.
Electrical test: The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.
Temperature rise tests: See table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.

TABLE I. Rubber handle boot.

Properties	
Color	Clear transparent
Specific gravity at 77°F	1.15 +/- 0.02
Chemical classification	ZZ-R-765
Shore A hardness	50+/- 5
Minimum tensile strength (lb/in 2/)	1100
Minimum elongation (percent)	450
Minimum tear strength (lb/in 2/)	150
Minimum bayshore resiliency (percent)	40
Fatigue life at 125 percent strength (flexes)	800,000
Test requirements	
Color	1/
Shore A hardness	ASTM D 2240
Tensile strength	ASTM D 412
Elongation	ASTM D 412
Tear strength	ASTM D 624
Bayshore resiliency	ASTM D 412
Fatigue life	ASTM D 412

1/ A 0.075 inch slab of the rubber shall be placed in direct contact with a piece of newsprint. The rubber shall not interfere with the ability to read the newsprint.

TABLE II. Instantaneous trip calibrations.

Type Designation	Trip element Rating	Test Current Amperes	Trip Time 1/ Seconds
AQB-A50 60 Hz, 400 Hz	10, 15, 20	30X (Element Rating)	.01 to .70
	25, 30	600	.01 to 1.30
	35, 40, 50	600	.01 to 3.00
AQB-A50 dc	10, 15, 20	40X (Element Rating)	.01 to .40
	25, 30	600	.01 to 1.10
	35, 40, 50	600	.01 to 2.50

1/ Test each pole individually; test current shall be preset and maintained within plus or minus 10 percent of specified value until breaker trips.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.

TABLE IIa. Instantaneous trip calibrations.

Type Designation	Trip element Rating	Test Current Amperes	Trip Condition 1/
AQB-A51	All Ratings	20X 40X	No Trip Trip

1/ Test each pole individually; test current shall be a current pulse of 80-100 milliseconds duration having no more than 5 percent current asymmetry in the first half cycle.

Preparing activity:
Navy - SH
(Project 5925-N165)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:00Z
Check the source to verify that this is the current version before use.